Computational Paralinguistics in Resource-Constrained Environments: Advanced Algorithms for "The Future You" Platform

Executive Summary

This research report presents a comprehensive technical upgrade for the "Result Analysis" engine of *The Future You* application. The current architecture, reliant on weighted averages, simple variance metrics, and static posture ratios, provides a foundational Minimum Viable Product (MVP) but lacks the nuance required for high-fidelity coaching. The "Logic Audit" reveals a system that is functional but mathematically naive; it captures the *existence* of communication signals (e.g., did the user smile?) but fails to capture the *dynamics* of those signals (e.g., was the smile synchronous with a positive statement, or did it lag, indicating masking?). To elevate the platform to a "PhD-level" standard without violating the zero-cost/local-AI constraint, we propose shifting from static scalar metrics to dynamic, vector-based analysis.

The proposed upgrades introduce four novel algorithmic frameworks designed specifically for the resource-constrained "Jugaad" stack (Node.js/Python on local CPU):

1. **Speech Coherence:** Replacing simple BERT variance with a **Simplified Entity Grid Model** and **Sliding-Window Semantic Progression**, allowing the system to distinguish between logical flow and mere topic repetition. This moves beyond "staying on topic" to measuring "narrative momentum."
2. **Vocal Energy & Confidence:** Implementing **Fundamental Frequency (F0) Modulation Analysis**, **Harmonics-to-Noise Ratio (HNR)**, and **Vowel Space Area (VSA)** to quantify "charisma," "clarity," and "articulation" using raw acoustic data processed via local Python libraries (e.g., librosa, parselmouth).
3. **Body Language & Presence:** Moving from static ratios to **Kinematic Energy Profiling** and **Convex Hull Openness Metrics** utilizing existing MediaPipe landmarks to score gesture dynamism, fluidity (jerk minimization), and postural authority.
4. **Emotional Intelligence:** Transitioning from single-label classification to **Multimodal Incongruence Detection**, flagging discrepancies between verbal sentiment, vocal tone, and facial expression as indicators of authenticity or anxiety.

These algorithms adhere to the project's zero-cost philosophy by utilizing mathematical transformations of open-source model outputs rather than relying on expensive generative AI APIs.

1. Introduction: The Shift to Computational Paralinguistics

The primary objective of *The Future You* is to gamify self-reflection and communication coaching. The current iteration of the "Logic Audit" operates on first-order metrics: counting filler words, averaging sentiment scores, and measuring static body ratios.[1] While computationally inexpensive, these metrics fail to capture the *temporal dynamics* of human communication—the rise and fall of pitch that signals charisma, the rhythmic cadence that indicates confidence, or the logical progression of entities that constitutes coherent thought.

This report outlines a transition to *Computational Paralinguistics*, a field that quantifies *how* things are said rather than just *what* is said.[2] By treating the user's video not as a static data point but as a time-series signal, we can extract second-order features (velocity, acceleration, variance, entropy) that correlate strongly with human perceptions of "confidence" and "competence."

1.1 The Limitations of Current Metrics (The "Average" Fallacy)

The current "Logic Audit" findings rely heavily on weighted averages. For example, a user might receive a "Clarity Score" based on the inverse frequency of "um" and "uh" combined with a pacing score. While useful, this linear approach has significant blind spots known as the "Average Fallacy":

- **False Positives in Coherence:** A speaker repeating the same sentence ten times would currently score high on "consistency" and low on "variance," effectively gaming the BERT-based variance check. The system mistakes repetition for stability.
- **Insensitivity to Micro-Expressions:** A static "neck-to-shoulder" ratio [1] ignores the fluid micro-adjustments a confident speaker makes. A stiff, frozen posture might score well mathematically but appear robotic or terrified to a human observer. The current system penalizes slouching but fails to reward dynamic engagement.
- **Tone Deafness via Aggregation:** Averaging sentiment scores over a whole transcript masks crucial narrative arcs. A speech that starts anxious but ends confident averages out to "neutral," missing the user's successful recovery and "hero's journey."

1.2 The "Zero-Cost" Constraints as Design Principles

The architectural blueprint strictly prohibits paid cloud services (AWS Transcribe, GPT-4), necessitating the use of open-source, offline models.[1] This constraint is not merely a limitation but a design directive to use *heuristics* and *feature engineering* over brute-force Large Language Models (LLMs).

- **Constraint:** Limited compute (laptop CPU).

- **Solution:** Efficient algorithms (e.g., Entity Grids, Signal Energy) that extract maximum insight from lightweight model outputs (e.g., MediaPipe landmarks, Vosk transcripts) rather than running heavy generative models. We prioritize algorithms with $O(N)$ or $O(N \log N)$ complexity over those requiring massive parallel processing.

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2. Advanced Speech Coherence: Structural Logic Algorithms

Coherence is the property of text that makes it semantically meaningful and logically connected. The current approach uses BERT embedding variance—calculating how much sentence vectors differ from one another.[1] The assumption is that low variance implies staying on topic. However, this is a flawed proxy. A disjointed list of synonyms has low variance but zero narrative coherence. Furthermore, highly engaging speeches often traverse different topics (high variance) but maintain a logical thread (high coherence).

We propose replacing simple variance with two advanced, lightweight algorithms: **The Simplified Entity Grid** for syntactic coherence and **Semantic Progression Vectors** for topical flow.

2.1 The Simplified Entity Grid Model (Local Implementation)

The Entity Grid Model, originally proposed by Barzilay and Lapata [5], is a gold-standard approach in computational linguistics for assessing local coherence. It posits that coherent texts exhibit specific patterns of entity transitions (e.g., a subject in one sentence becoming the object in the next).

2.1.1 Theoretical Framework: The Matrix of Focus

In a coherent discourse, entities (people, places, concepts) are introduced and then referred back to. The "Grid" is a matrix where rows correspond to sentences ($S_1, S_2,... S_n$) and columns correspond to entities ($E_1, E_2,... E_m$).
Each cell $C_{i,j}$ contains the grammatical role of entity j in sentence i:

- **S:** Subject
- **O:** Object
- **X:** Other argument
- **-:** Absent

Coherence is measured by the probability of transition sequences (e.g., $S \rightarrow O$, $O \rightarrow -$, $S \rightarrow S$).[6] Coherent texts tend to have dense columns with frequent $S \rightarrow S$ or $S \rightarrow O$ transitions (maintaining focus), whereas incoherent texts have sparse columns or chaotic transitions.

2.1.2 Zero-Cost Implementation Strategy

Implementing a full dependency parser to distinguish Subject/Object can be computationally heavy for a local CPU stack. To adhere to the "Jugaad" philosophy [1], we propose a **Simplified Entity Grid** using shallow parsing via spacy (which has optimized, lightweight models like en_core_web_sm) or even compromise (a very fast NLP library for Node.js).

Algorithm 1: Local Coherence Scoring via Simplified Entity Transitions

1. **Entity Extraction:** Use a lightweight Named Entity Recognition (NER) and Noun Phrase extraction on the transcript.
   - *Input:* "I started the project. It was difficult. The team helped me."
   - *Entities:* "I", "Project", "Team".
2. **Grid Construction:** Construct a boolean matrix where 1 indicates presence and 0 indicates absence (skipping the complex S/O/X roles to save compute).
   - *Row 1:* {I: 1, Project: 1, Team: 0}
   - *Row 2:* {I: 0, Project: 1 (via 'It'), Team: 0}
   - *Row 3:* {I: 1, Project: 0, Team: 1}
3. Transition Weighting: Calculate a weighted score based on the distance between re-mentions.
   Let $d(e)$ be the distance in sentences between consecutive mentions of entity e.

   $$Score_{coherence} = \frac{1}{N_E} \sum_{e \in Entities} \frac{1}{1 + \ln(\bar{d}(e))}$$

   Where N_E is the total number of unique entities and $\bar{d}(e)$ is the average distance between mentions of entity e.

Rationale: This formula rewards texts where entities are reintroduced relatively quickly (low distance), implying a tight narrative thread. It penalizes "word salad" where new entities are constantly introduced but never followed up on.

2.2 Semantic Progression Vectors (Sliding Window Cosine Similarity)

While Entity Grids measure *syntactic* cohesion, we must also measure *semantic* flow. We retain the BERT embeddings but change the mathematical approach from *global variance* to *local progression*.[7]

2.2.1 The Cosine Similarity Chain

Instead of calculating the variance of all sentence vectors against a global mean (which penalizes a speech that evolves from Topic A to Topic B), we calculate the **cosine similarity between adjacent sentences**.[9]

Let V_i be the BERT embedding of sentence i. We calculate the similarity vector C:

$$C_i = \text{CosineSimilarity}(V_i, V_{i+1}) = \frac{V_i \cdot V_{i+1}}{\|V_i\| \|V_{i+1}\|}$$

2.2.2 The "Smoothness" Metric (1st Order Derivative)

A coherent speech should have smooth transitions. Abrupt drops in C_i indicate non-sequiturs. We define the **Semantic Flow Score (S_{flow})**:

$$S_{flow} = 100 \times \left(1 - \frac{\sum_{i=1}^{n-1} \max(0, T - C_i)}{n-1} \right)$$

- T is a threshold constant (empirically set around 0.5 to 0.6 for BERT embeddings).
- The term $\max(0, T - C_i)$ penalizes only those transitions where similarity drops below the threshold (disjointed sentences).
- Smooth transitions ($C_i > T$) incur no penalty.

Insight: This algorithm allows for "Topic Evolution." If a speaker slowly transitions from "My childhood" to "My career," adjacent sentences will remain similar ($C_i \approx 0.8$), maintaining a high score. A global variance model would incorrectly penalize this evolution as "high variance."

2.2.3 Third-Order Insight: Connective Density

To further refine coherence, we can perform a keyword scan for **Discourse Markers** (e.g., "however," "therefore," "consequently," "firstly").

- *Algorithm:* Count(DiscourseMarkers) / TotalSentences.
- *Interpretation:* High density of discourse markers usually correlates with high perceived logic, as these words explicitly signal the relationship between ideas.[10] This serves as a low-compute "sanity check" on the BERT-based smoothness score.

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3. Vocal Energy & Confidence: Acoustic Feature Engineering

The current "Energy" score likely uses RMS (Root Mean Square) amplitude, effectively measuring loudness.[1] However, loudness is not confidence. A speaker can be loud and monotonous (boring) or quiet and intense (captivating). "Charisma" in speech is acoustically defined by specific patterns of **Pitch (F0) Dynamics** and **Spectral Clarity** (HNR).[11]

To upgrade this, we introduce algorithms utilizing Python libraries like librosa or parselmouth (wrappers for Praat).[12]

3.1 Fundamental Frequency (F_0) Analysis

The fundamental frequency ($F0$) corresponds to the perceived pitch of the voice. Research

in charismatic speech indicates that **F0 variability** (standard deviation) is a strong predictor of perceived charisma and confidence.[2] Monotone speakers appear lethargic or unsure; dynamic speakers appear engaging.

3.1.1 The Physics of "Boring" vs. "Dynamic"

Human attention is driven by change. A flat pitch line ($F_0(t) \approx k$) causes listener habituation (boredom). A modulated pitch line signals information importance.

- **Micro-prosody:** Jitter (cycle-to-cycle variation). High jitter = Pathological/Nervous.
- **Macro-prosody:** Intonation (sentence-level variation). High intonation = Charismatic.
- **Challenge:** Simple standard deviation captures both. We must distinguish them.

3.1.2 Algorithm: The Dynamic Range Score (Semitone Normalized)

1. **F0 Extraction:** Use the librosa.pyin (Probabilistic YIN) algorithm. It is robust to noise and runs on CPU.[15]
    - *Input:* Audio waveform.
    - *Output:* Time series of $F0$ values (Hz).
2. **Voiced Frame Filtering:** Discard unvoiced frames (silence/noise) where probability < 0.5.
3. Semitone Conversion: Linear Hz scales are misleading because human pitch perception is logarithmic. A 20Hz jump is massive for a deep voice (80Hz) but negligible for a high voice (250Hz). We normalize to Semitones (st) relative to the speaker's mean:

    $$F0_{st}(t) = 12 \times \log_2 \left(\frac{F0(t)}{\overline{F0}} \right)$$
4. **Standard Deviation (σ_{st}):** Calculate the standard deviation of the semitone values.
    - *Thresholds:*
        - $\sigma_{st} < 1.5$: Monotone (Low Confidence).
        - $1.5 \le \sigma_{st} \le 2.5$: Normal conversational.
        - $\sigma_{st} > 2.5$: Dynamic/Charismatic (High Confidence).

Mathematical Improvement: Normalizing to semitones ensures the metric works equally well for male and female voices, removing gender bias from the confidence score.[16]

3.2 Harmonics-to-Noise Ratio (HNR)

HNR quantifies the purity of the voice signal. It measures the ratio of energy in the periodic (harmonic) components to the energy in the aperiodic (noise) components.[11]

- **High HNR (> 20dB):** Clear, resonant, confident voice. Indicates efficient vocal fold closure.
- **Low HNR (< 10dB):** Breathy, hoarse, or hesitant voice. Indicates air leakage, often correlated with lack of confidence or anxiety.

3.2.1 Calculation via Parselmouth

Using parselmouth (Python), HNR is calculated directly from the sound object via autocorrelation.

$$HNR_{dB} = 10 \log_{10} \left(\frac{E_{harmonics}}{E_{noise}} \right)$$
Integration Logic:
If $HNR < 12 \text{dB}$, flag as "Breathy/Unsure."
If $HNR > 20 \text{dB}$, reward as "Resonant/Clear".17

3.3 Articulation Clarity: Vowel Space Area (VSA)

Articulation rate (words per minute) is currently tracked, but clarity of pronunciation is not. A fast speaker can still be unintelligible.
Proposed Metric: Vowel Space Area (VSA).

- **Concept:** Clear speakers articulate vowels distinctly (High F1/F2 separation). Mumbling speakers centralize vowels (Low separation).
- **Algorithm:**
  1. Extract Formants (F1, F2) using parselmouth.Sound.to_formant_burg().
  2. Identify the extreme points for /i/, /a/, /u/ (corner vowels).
  3. Calculate the area of the triangle formed by these points in the F1-F2 plane.
  4. Higher Area = Better Articulation.

3.4 The "Vocal Authority" Weighted Metric

We combine these features into a single composite metric to replace the simple "Energy" score.

$$Score_{vocal} = w_1 \cdot \sigma_{st}^{norm} + w_2 \cdot HNR_{norm} + w_3 \cdot Intensity_{range}$$

- σ_{st}^{norm}: Normalized Pitch Variability (0-100).
- HNR_{norm}: Normalized Voice Quality (0-100).
- $Intensity_{range}$: The difference between the 90th and 10th percentile of loudness (dynamic range), preventing "shouting" from scoring high.

Suggested Weights: $w_1=0.5$ (Pitch is dominant for charisma), $w_2=0.3$ (Quality), $w_3=0.2$ (Volume dynamics).

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4. Body Language & Presence: Kinematic Metrics via MediaPipe

The current implementation uses simple ratios like neck_length / shoulder_width to detect slouching.[1] This is static and brittle. It fails to account for camera distance (z-axis) and completely ignores the *movement* (gestures) that defines public speaking presence.

We utilize **MediaPipe Pose**, which provides 33 3D landmarks (x, y, z). We propose upgrading to **Kinematic Energy** and **Convex Hull Openness**.

4.1 The Openness Vector (Postural Dominance)

Psychological research (e.g., Cuddy et al.) suggests "power posing" or expansive postures correlate with confidence.[18] We can quantify this mathematically using the geometry of the shoulders and torso.

4.1.1 Algorithm: The Torso Expansion Index (Convex Hull)

Instead of a simple ratio, we calculate the area of the upper torso polygon.
Let L_{11} (Left Shoulder), L_{12} (Right Shoulder), and L_{0} (Nose) be landmark coordinates (x,y,z).

1. **Shoulder Width ($W_{shoulders}$):** Euclidean distance between L_{11} and L_{12}.
2. Triangle Area (A_{torso}): Area of the triangle formed by L_{11}, L_{12}, L_{0}.

   $$A_{torso} = \frac{1}{2} |x_{11}(y_{12} - y_{0}) + x_{12}(y_{0} - y_{11}) + x_{0}(y_{11} - y_{12})|$$
3. **Normalization:** Normalize A_{torso} by the bounding box of the user to account for camera distance.
4. **Z-Axis Slouch Detection:** Use the z coordinate. Ideally, shoulders should be vertically aligned with the ears.
   - $\Delta Z = (L_{11}.z + L_{12}.z)/2 - (L_{7}.z + L_{8}.z)/2$. (Shoulders vs Ears).
   - If $\Delta Z > Threshold$, the shoulders are rolled forward (slouching).

Logic:

$$Score_{posture} = \frac{A_{torso}}{A_{max_observed}} \times (1 - \text{SlouchFactor})$$

We track the maximum expansion observed during the user's calibration phase and score frames relative to their own best posture.

4.2 Gesture Energy and Fluidity

Charismatic speakers use gestures to emphasize points. Static hands indicate nervousness; flailing hands indicate loss of control. We need a metric for "Controlled Energy".[19]

4.2.1 Algorithm: Kinetic Energy of the Wrists

1. **Track Landmarks:** Left Wrist (L_{15}) and Right Wrist (L_{16}).
2. Calculate Velocity (v):

   $$v_t = \sqrt{(x_t - x_{t-1})^2 + (y_t - y_{t-1})^2}$$

   (Calculated per frame t).
3. Calculate Gesture Energy ($E_{gesture}$):

   $$E_{gesture} = \sum_{t=1}^{N} (v_{t, left}^2 + v_{t, right}^2)$$
4. Jerk Minimization (Fluidity): Jerk is the derivative of acceleration. High jerk appears "spasmodic" or nervous. Low jerk appears "fluid" and confident.

   $$Jerk = \frac{d^3x}{dt^3}$$

   Lower average Jerk score = Higher "Grace/Confidence" score.

Interpretation:

- **High Energy + Low Jerk:** Passionate/Persuasive (The ideal state).
- **High Energy + High Jerk:** Chaotic/Nervous (Flailing).
- **Low Energy:** Passive/Disengaged.

4.2.2 Second-Order Insight: Gesture-Speech Synchrony

We can check if peaks in Gesture Energy align with peaks in Vocal Intensity.

- **Synchrony:** If $Peak(E_{gesture})$ occurs within $\pm 200ms$ of $Peak(Intensity_{vocal})$, it indicates high emotional authenticity.[20]
- **Asynchrony:** Gestures that lag speech indicate "cognitive load" or rehearsed/fake acting.

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5. Emotional Intelligence: Multimodal Incongruence Detection

Current sentiment analysis classifies text as "Happy" or "Sad".[1] This is insufficient. A user saying "I am so happy to be here" with a flat pitch and slumped shoulders is not happy; they are bored or sarcastic. This phenomenon is known as **Emotional Leakage**.

We propose **Multimodal Incongruence Detection**. This does not require training a massive multimodal model (like GPT-4o). It uses *late fusion* of the metrics we have already calculated.

5.1 The Incongruence Algorithm

We define a vector state for the user at time t.

State $S_t =$

1. **Normalize Features:** Map all features to a -1 (Negative/Low) to +1 (Positive/High) scale.
    - *Text:* Sentiment score (-1 to 1).
    - *Audio:* Pitch relative to mean (Low = -1, High = 1).
    - *Body:* Posture Openness (Closed = -1, Open = 1).
2. Calculate Divergence:
   We calculate the standard deviation or variance between these three modes at any given segment.

   $$D_{incongruence} = \text{Variance}(Text_{val}, Audio_{pitch}, Body_{open})$$

5.2 Scoring Rule

- **Low Divergence:** The user is authentic (e.g., Sad text + Quiet voice + Closed posture = Authentic Sadness).
- **High Divergence:** The user is masking or incongruent (e.g., "Excited" text + Monotone voice).

Feedback Generation:
Instead of just "You sound sad," the AI can say: "Your words were positive, but your vocal energy didn't match. This creates a perception of insincerity. Try increasing your pitch variety to align with your message." This is PhD-level coaching advice derived from simple mathematical variance.

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6. Implementation Strategy: The Zero-Cost Architecture

To ensure these advanced algorithms run within the "Zero-Cost/Local" constraints [1], we must optimize the compute pipeline.

6.1 The Polyglot Worker Pattern

The current stack uses Node.js. While Node is excellent for orchestration, it is poor for heavy matrix math (Entity Grids, FFTs).

- **Proposal:** Use Node.js child_process to spawn ephemeral **Python scripts** only for the heavy lifting.
- **Libraries:**
    - *Python:* numpy (math), librosa (audio), spacy (NLP), mediapipe (vision).
    - *Communication:* Node sends file paths or JSON buffers to Python; Python returns a lightweight JSON analysis object.
    - *Latency:* Python script startup time can be mitigated by keeping a persistent Python shell open (using python-shell npm package) or by using lightweight C++ bindings if

available (though Python is easier for MVP).

6.2 Optimization Techniques for Laptop CPUs

1. **Downsampling:** MediaPipe does not need to run at 30 FPS. Posture changes slowly. Process every 5th frame (6 FPS) to reduce CPU load by 80% without losing meaningful signal quality.
2. **Audio Chunking:** Process audio in 10-second chunks rather than loading the whole file into RAM. This prevents memory overflows on standard laptops (4GB/8GB RAM constraint).
3. **Lazy Loading:** Do not load the BERT model until the video upload is complete. Unload it immediately after inference to free up resources for the React frontend.
4. **Quantization:** Use quantized versions of BERT and Vosk models (int8) to reduce memory footprint by 4x with negligible accuracy loss.

6.3 Data Structure Update

The Result Analysis JSON object in Supabase must be updated to store time-series data to enable the frontend to render "Confidence over Time" graphs.

JSON

```
{
  "speech_analysis": {
    "coherence_score": 85,
    "semantic_progression": [0.8, 0.85, 0.4, 0.9], // For graphing topic drops
    "entity_density": 0.45
  },
  "vocal_analysis": {
    "f0_std_semitones": 2.3,
    "hnr_db": 18.5,
    "dynamic_range_score": 78
  },
  "kinematics": {
    "posture_openness_avg": 0.88,
    "gesture_energy_peaks": 12, // Count of emphatic gestures
    "nervous_fidgeting_index": 0.1 // High freq, low amplitude movement
  }
}
```

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7. The Unified "Future You" Scoring Algorithm

Finally, we replace the linear weighted average with a non-linear, asymptotic scoring system. Linear scores are bad because they allow a user to "max out" one metric (e.g., shouting for high volume) to compensate for poor performance elsewhere.

7.1 The Sigmoid Normalization Function

We map raw metrics (like F0 variance or Gesture Energy) to a 0-100 score using a logistic function.[21] This provides a "soft cap" and diminishing returns for extreme values.

$$Score(x) = \frac{100}{1 + e^{-k(x - x_{target})}}$$

- x: Raw metric value.
- x_{target}: The ideal value (e.g., 2.5 st for pitch variability).
- k: Steepness curve (sensitivity).

7.2 The "Charisma Quotient" (Geometric Mean)

Instead of an arithmetic mean ($A+B+C+D)/4$), we use a **Geometric Mean** for the final score:

$$Score_{final} = \sqrt{S_{speech} \times S_{vocal} \times S_{body} \times S_{emotion}}$$
Why? The Geometric Mean penalizes imbalance. If *any* single category is 0 (e.g., zero vocal energy), the entire score collapses toward zero. This forces the user to be well-rounded. A speaker with perfect text but robot voice will score lower here than in a weighted average, accurately reflecting that they are a poor speaker.

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8. Conclusion

By implementing these upgrades, *The Future You* moves from a simple "checker" to a sophisticated "analyzer."

1. **Simplified Entity Grids** provide deep insight into thought structure without needing GPT-4.
2. **Acoustic F0/HNR analysis** gives professional-level feedback on vocal quality.
3. **Kinematic Energy** metrics turn simple video frames into physics-based presence scores.
4. **Multimodal Fusion** detects the nuance of authenticity.

This architecture respects the zero-cost constraint while delivering feedback comparable to

high-end, cloud-based coaching platforms. It represents a shift from "detecting what is wrong" to "analyzing the dynamics of what makes communication right."

Comparison with Old vs. New Metrics:

Feature	Old Logic (Current)	PhD-Level Upgrade (Proposed)
Coherence	BERT Variance (prone to errors)	Entity Grid + Semantic Progression
Confidence	Posture Ratio (static)	Openness Hull + Gesture Energy
Voice	RMS Volume (loudness)	F0 Modulation + HNR (Charisma)
Emotion	Text Sentiment (unimodal)	Multimodal Incongruence (Authenticity)
Scoring	Linear Average (hackable)	Geometric Mean + Sigmoid (Robust)

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